



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

Digital Forensics



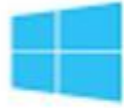
Linux for Digital Forensics

Overview

- Introduction to OS and file Systems
- Virtual file system
- File structure
- Path and path variable
- Linux commands



Android



Windows 10



Microsoft
Windows



Ubuntu



Linux

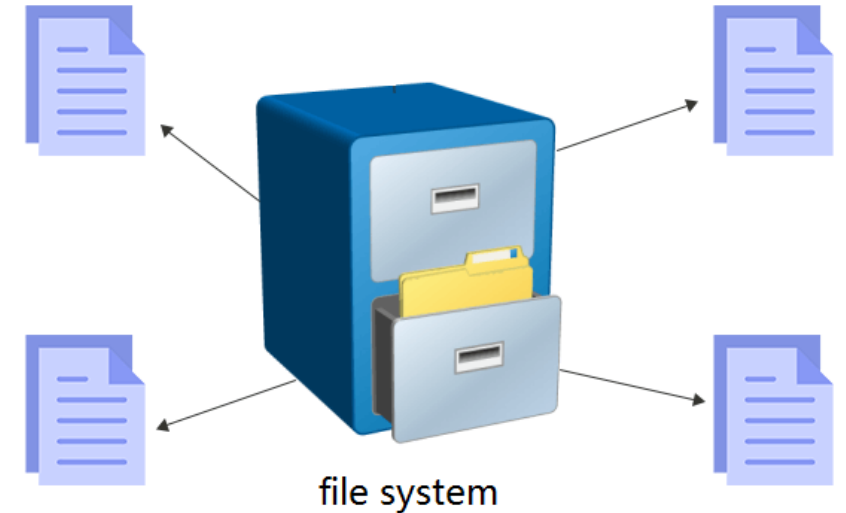


macOS

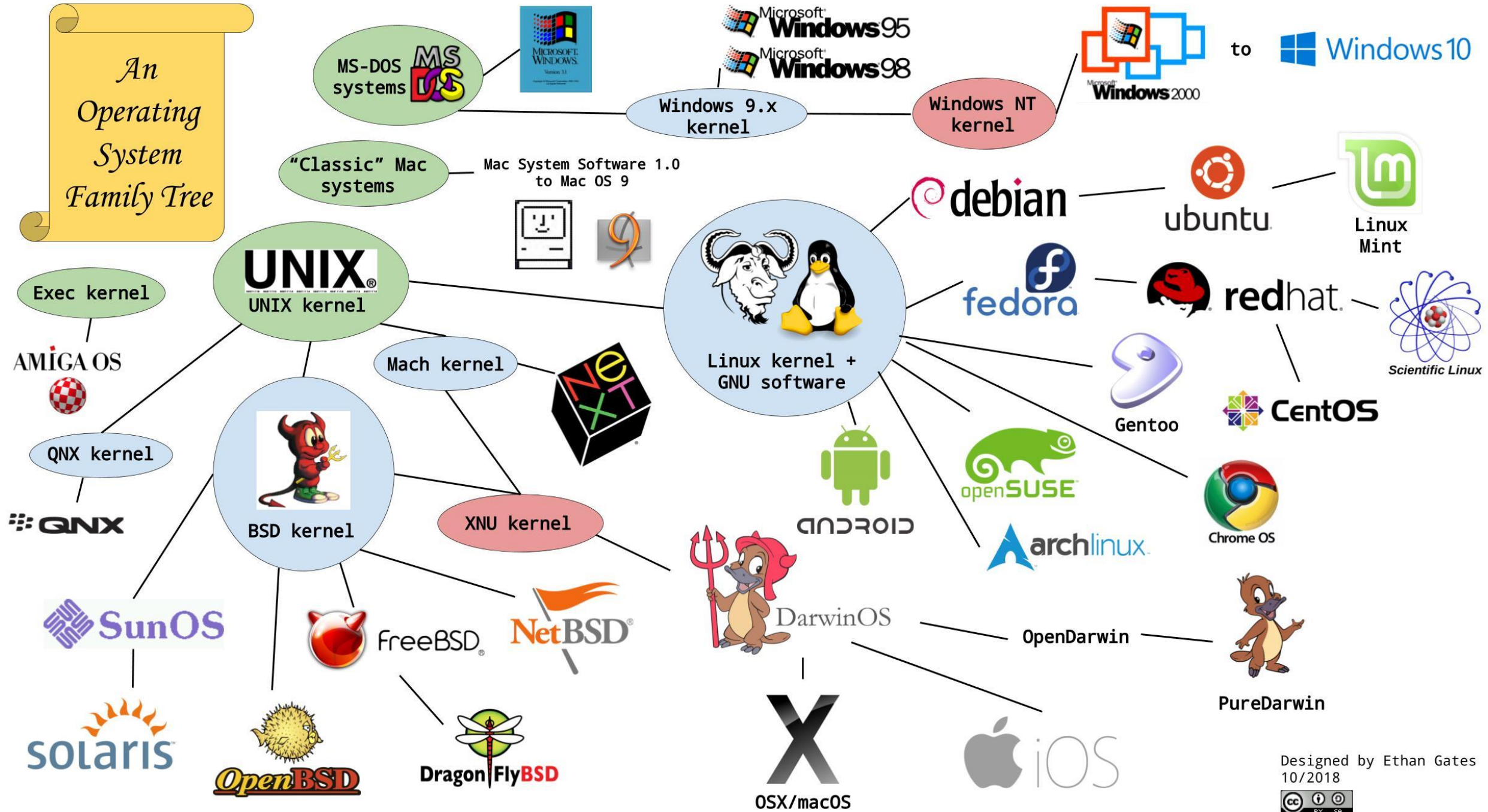


Chrome OS

OS and File Systems

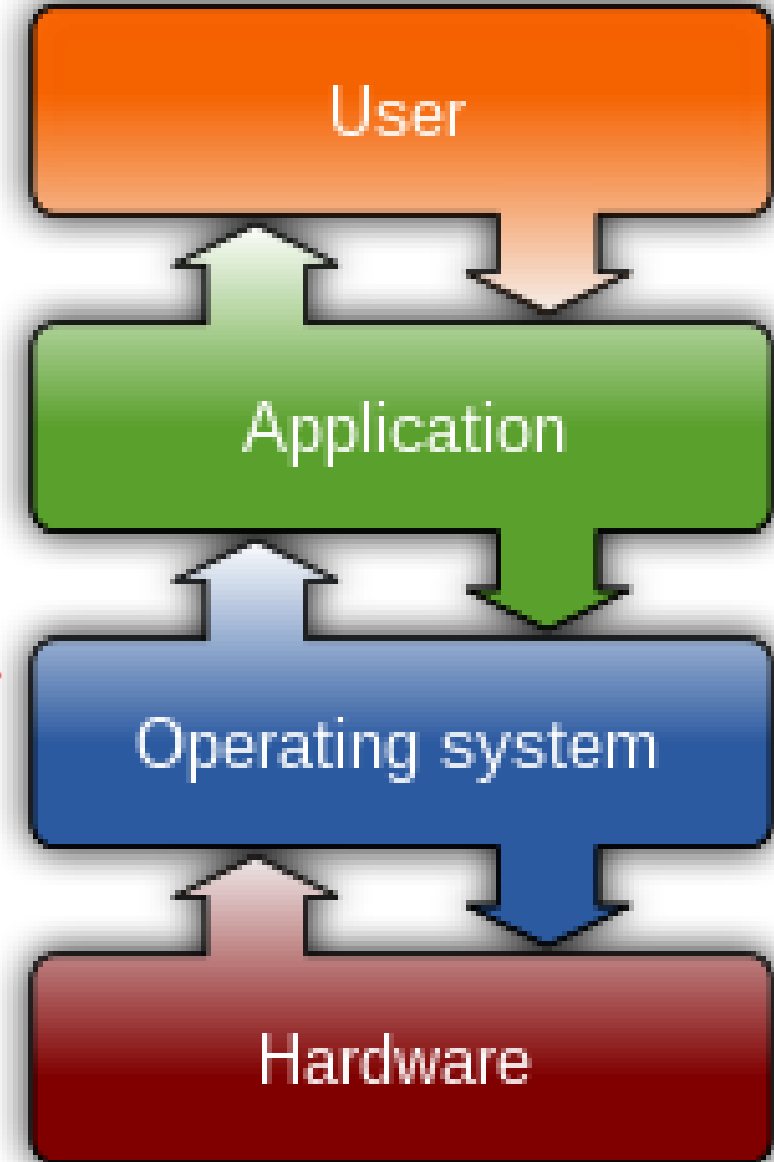


An Operating System Family Tree



Common features of OS

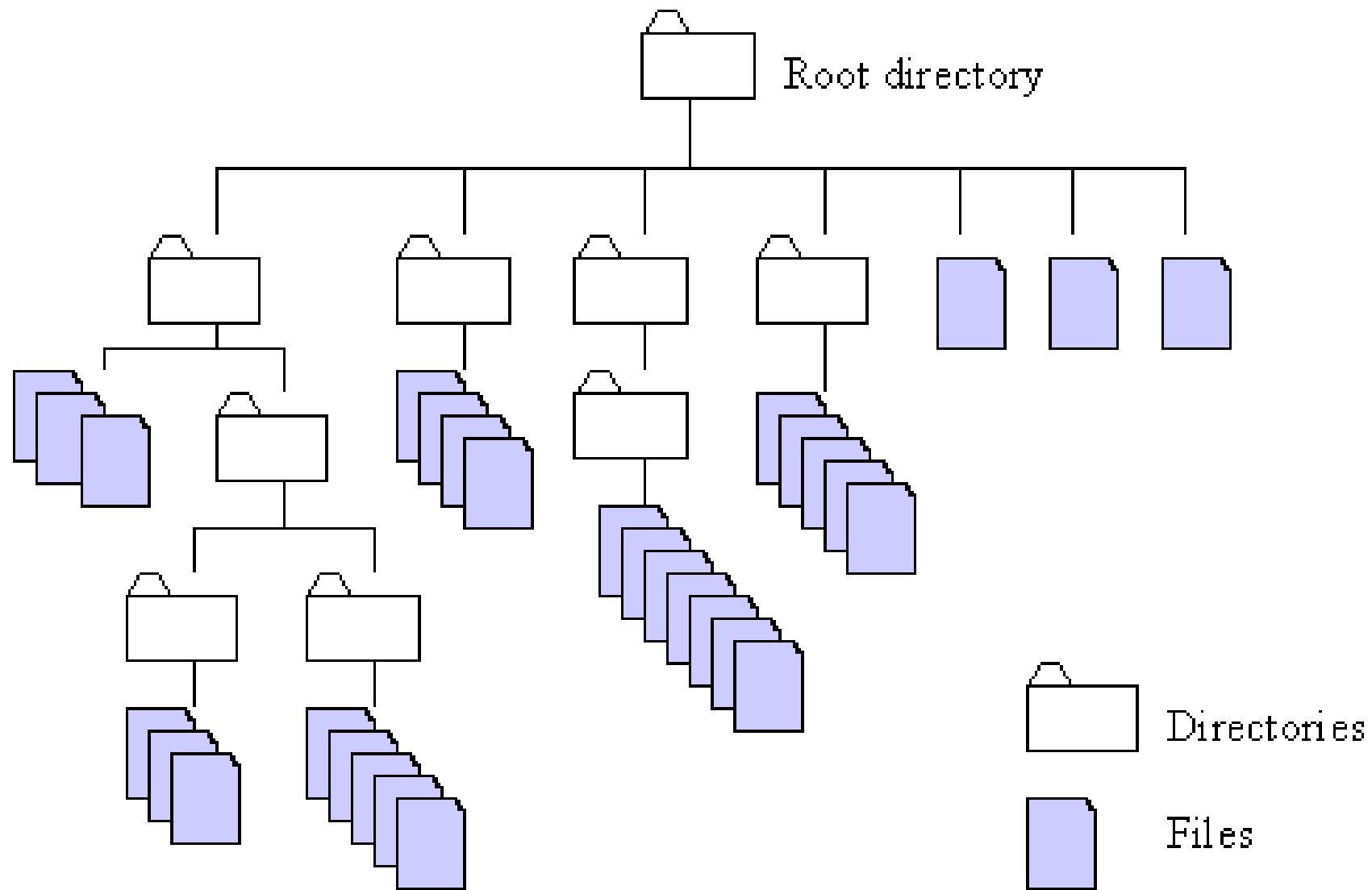
- Process management
- Memory management
- File system
- Device drivers
- Networking
- Security
- I/O



File Systems

- A file
 - is a collection of correlated information
 - information is recorded on secondary or non-volatile storage like magnetic disks, optical disks, and tapes.
- A file system
 - defines how files are named, stored, and retrieved from a storage device.





File systems used by operating systems

- By Unix and Unix-like operating systems

- Linux: XFS, JFS, and btrfs.

- Solaris

- macOS:

- Hierarchical File System (HFS) + : No support for dates beyond February 6, 2040

- By Microsoft Windows

- FAT: File Allocation Table
 - NTFS: New Technology File System

Ext24
extended file system

APFS
Apple File System



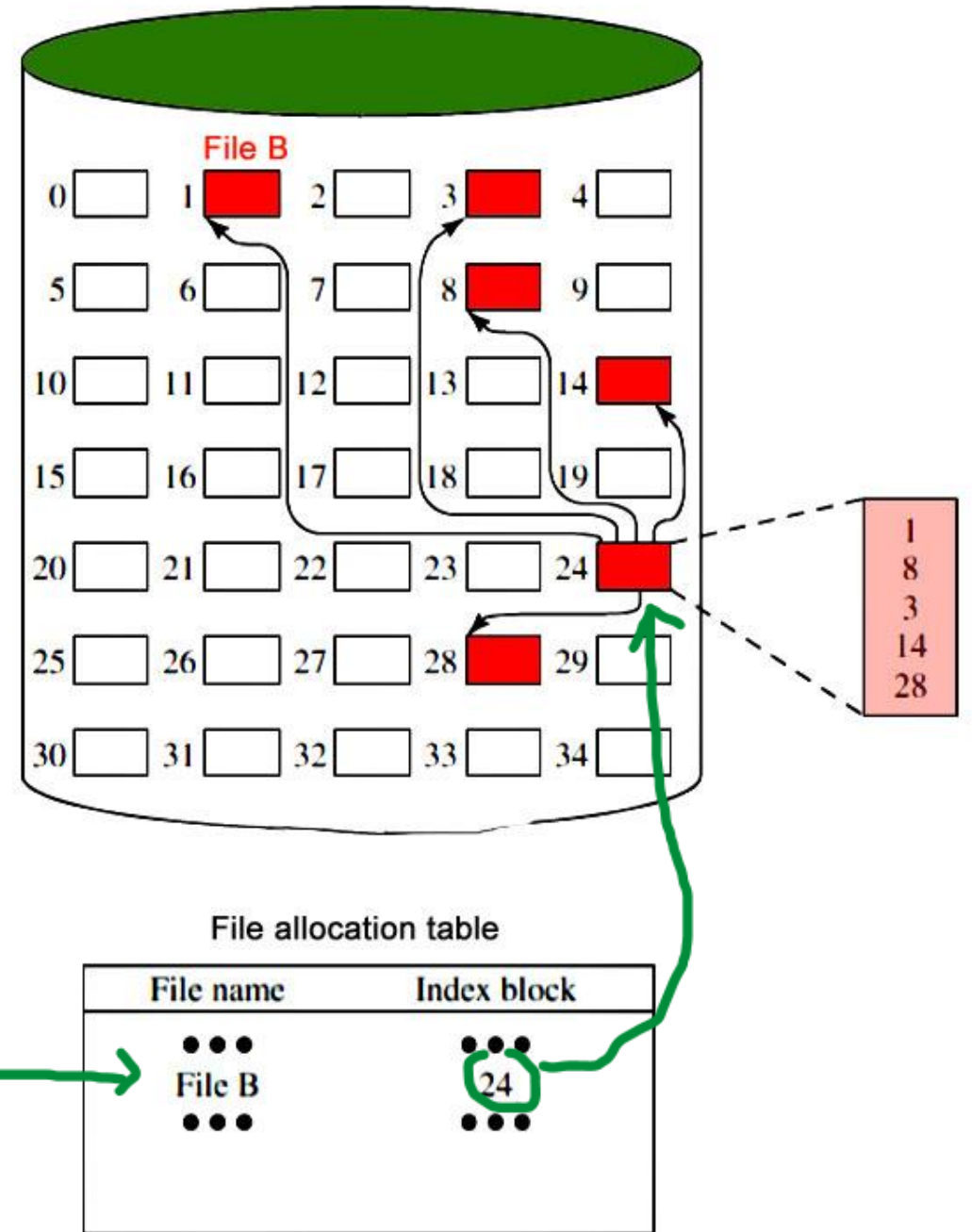

Microsoft

Manage files vs. manage storage boxes

Assumption: One person rents one or multiple boxes.

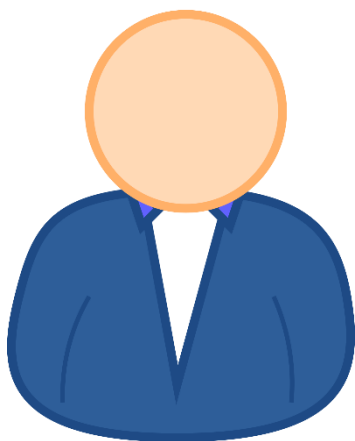
Name	Boxes ID
Frank	1,2,3
...	...



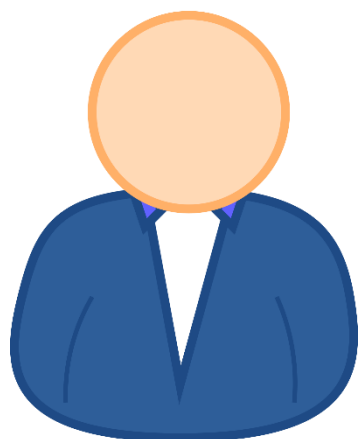


	Storage	File Storage
Unit	Box	Sector
Usage	Person rents boxes	File uses sectors
Management system	Rental management system	File management system, (e.g., FAT)

Virtual file system (VFS)

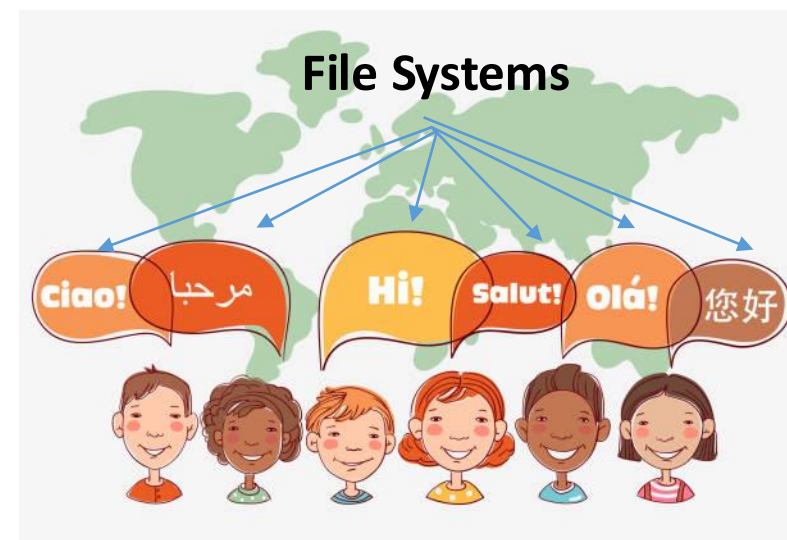


Without an interpreter



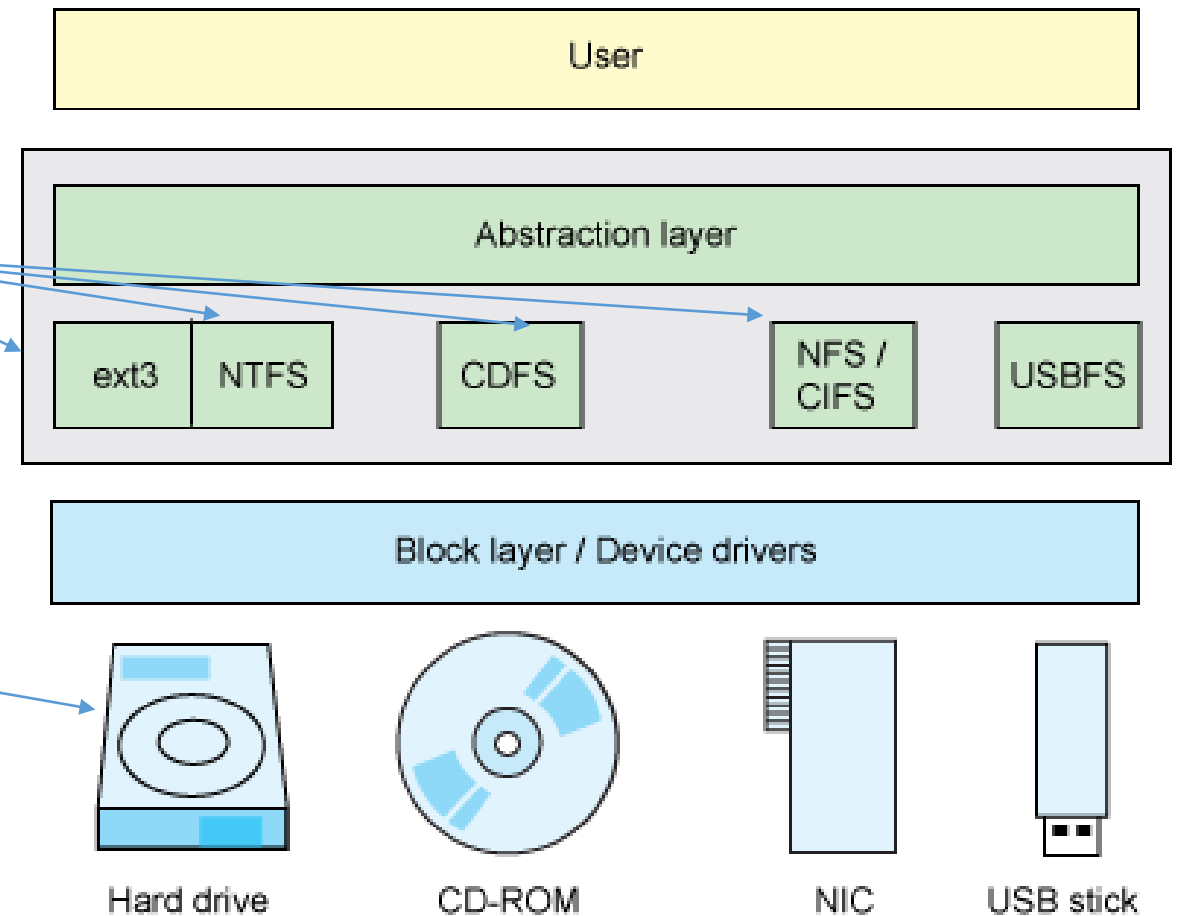
With an interpreter

Virtual file system



Linux is a VFS

- VFS allows client applications to access different types of **concrete** file systems in a uniform way
 - provides an abstract layer for upper-layer applications
 - The same techniques can be utilized to investigate different types of devices
- Each and everything in Linux is a **file** (Everything appears somewhere in the filesystem)
 - file, directory, hard disks, CD/DVD, NIC, USB
 - devices can be represented as file-like objects under `/dev/` filesystem.
- OS recognizes files by
 - inode (index node)



Linux for digital forensics- Good and Bad

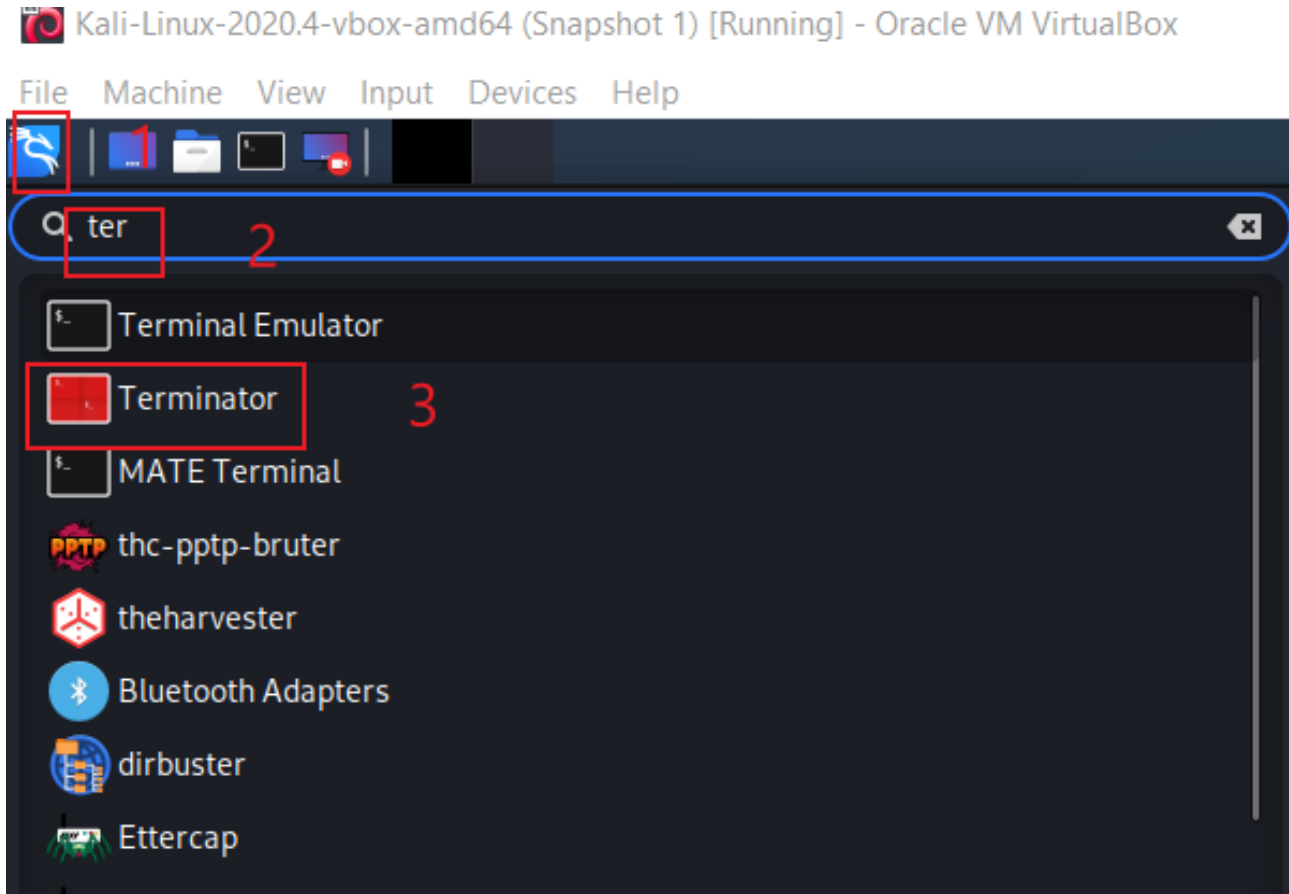
- Pro

- widely support for many file systems (Virtual file system)
- advanced and powerful tools available
- free and open-source
- compile source code
- flexibility

- Con

- learning curve, e.g., no graphic interface
- trustworthiness of open-source tools

Open a Linux Terminator



same as cmd

File inode



dir

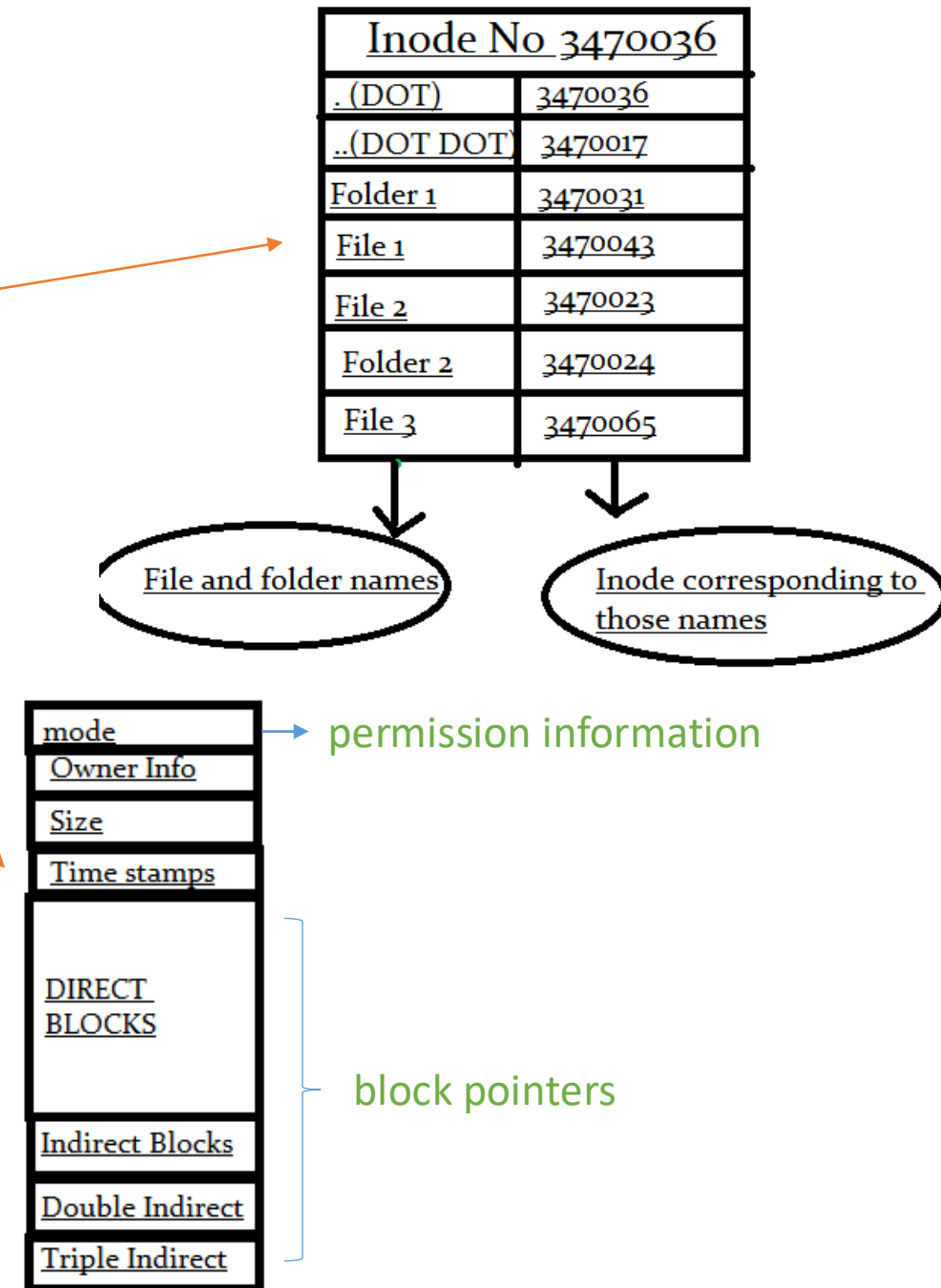
```
(kali㉿kali)-[~]
$ ls
Desktop  Downloads  Marvic_air  Pictures  Public  UB-730-Tools
Documents  hacking_case  Music  pixel3  Templates  Videos

(kali㉿kali)-[~]
$ ls -la
401164 .
393217 ..
401192 .bash_history
401174 .bash_logout
401170 .bashrc
401176 .bashrc.original
531550 .cache
401180 .config
531551 Desktop
401178 .dmrc
660979 Documents
660975 Downloads
401173 .face
401175 .face.icon
401177 .gnupg
401370 hacking_case
401198 .ICEauthority
401472 .java
660983 .local
401498 Marvic_air
401206 .mozilla
660980 Music
660981 Pictures
796226 pixel3
401171 .profile
660978 Public
401468 .razorsql
401455 .rs
401456 .sqlite_history
660977 Templates
661438 UB-730-Tools
401193 .vboxclient-clipboard.pid
401200 .vboxclient-display-svga-x11.pid
401199 .vboxclient-draganddrop.pid
401195 .vboxclient-seamless.pid
660982 Videos
402204 .viminfo
401203 .wget-hsts
401187 .Xauthority
401188 .xsession-errors
401191 .xsession-errors.old
401369 .zsh_aliases
402209 .zsh_history
401172 .zshrc
```

- **-i, --inode** print the index number of each file
- **-a, --all** do not ignore entries starting with . (hidden file)

What is inode?

- inode is an ID of a file/folder
 - unique number
- inode is a data structure of a file
 - You store your information in a file, and the operating system stores the information/metadata about a file in an inode
- inode points to the content of a file
- How to access to a file?
 - inode ID
 - file name=>inode ID



```
(kali㉿kali)-[~]
```

```
$ ls --help
```

Usage: ls [OPTION]... [FILE]...

List information about the FILES (the current directory by default).

Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.

-a, --all	do not ignore entries starting with .
-A, --almost-all	do not list implied . and ..
--author	with -l, print the author of each file
-b, --escape	print C-style escapes for nongraphic characters
--block-size=SIZE	with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below
-B, --ignore-backups	do not list implied entries ending with ~
-c	with -lt: sort by, and show, ctime (time of last modification of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first
-C	list entries by columns
--color[=WHEN]	colorize the output; WHEN can be 'always' (default if omitted), 'auto', or 'never'; more info below
-d, --directory	list directories themselves, not their contents
-D, --dired	generate output designed for Emacs' dired mode
-f	do not sort, enable -aU, disable -ls --color
-F, --classify	append indicator (one of */=>@) to entries
--file-type	likewise, except do not append '*'
--format=WORD	across -x, commas -m, horizontal -x, long -l, single-column -1, verbose -l, vertical -C
--full-time	like -l --time-style=full-iso
-g	like -l, but do not list owner
--group-directories-first	group directories before files

Disk free information (*df*) inode

```
kali@kali: ~ 66x20

(kali@kali)-[~]
$ df -i
```

-i, --inodes list inode information

Filesystem	Inodes	IUsed	IFree	IUse%	Mounted on
udev	5645019	424	5644595	1%	/dev
tmpfs	5653432	656	5652776	1%	/run
/dev/sda1	5185536	417763	4767773	9%	/
tmpfs	5653432	1	5653431	1%	/dev/shm
tmpfs	5653432	2	5653430	1%	/run/lock
tmpfs	1024	17	1007	2%	/sys/fs/cgroup
tmpfs	1130686	63	1130623	1%	/run/user/1000



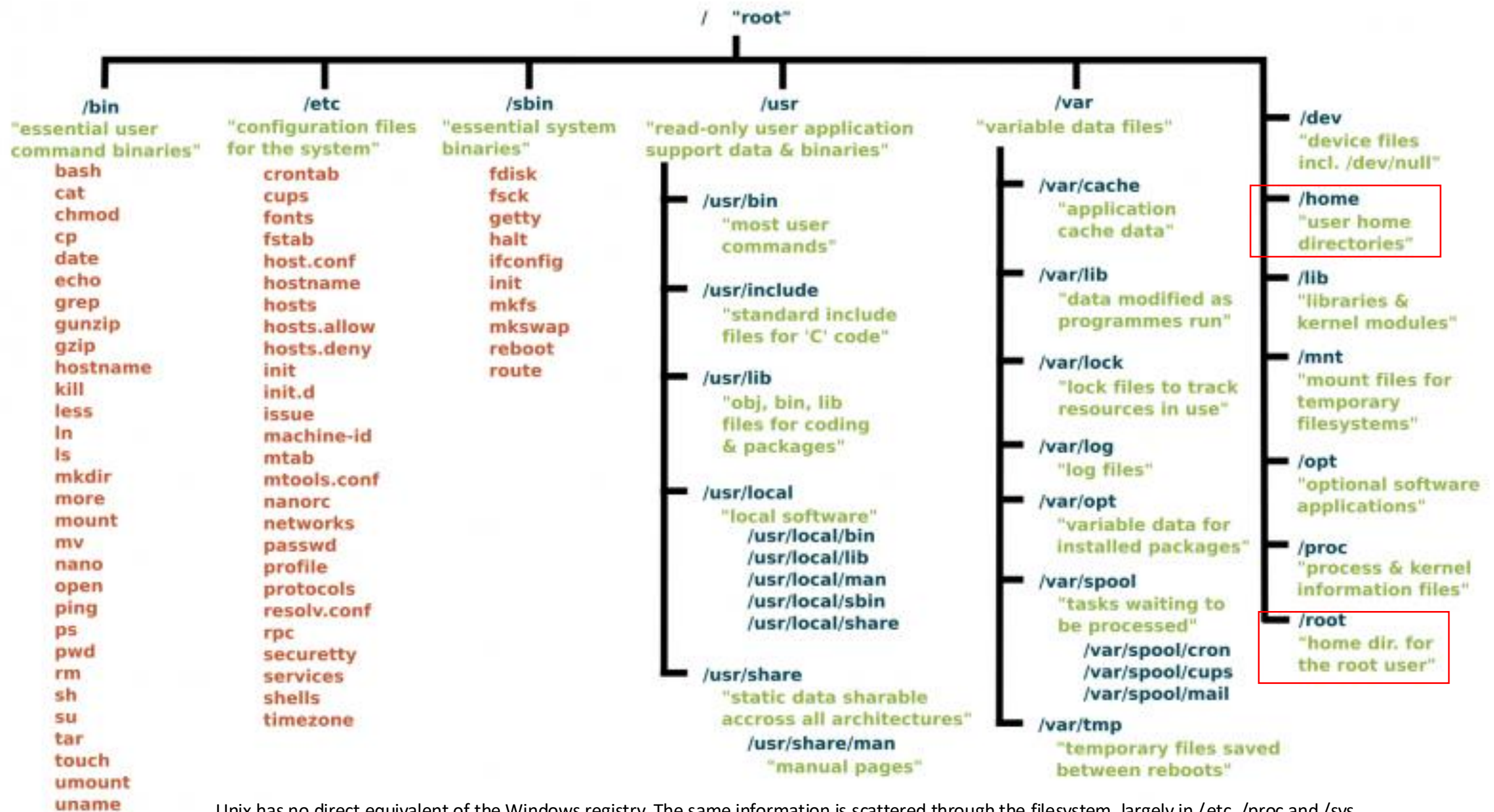
Show human readable format

```
(kali㉿kali)-[~]
```

```
$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	22G	0	22G	0%	/dev
tmpfs	4.4G	964K	4.4G	1%	/run
/dev/sda1	78G	52G	23G	70%	/
tmpfs	22G	0	22G	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
tmpfs	4.0M	0	4.0M	0%	/sys/fs/cgroup
tmpfs	4.4G	52K	4.4G	1%	/run/user/1000

File Structure



Unix has no direct equivalent of the Windows registry. The same information is scattered through the filesystem, largely in /etc, /proc and /sys.

Trash



File System



Home



RazorSQL



Seagate3T



157 GB
Volume

File System - File Manager

File Edit View Go Help

← → ↑ ↗ 📁 /

DEVICES

📁 File System

📁 157 GB Volume

📁 Seagate3T

PLACES

📁 kali

📁 Desktop

📁 Trash

📁 Documents

📁 Music

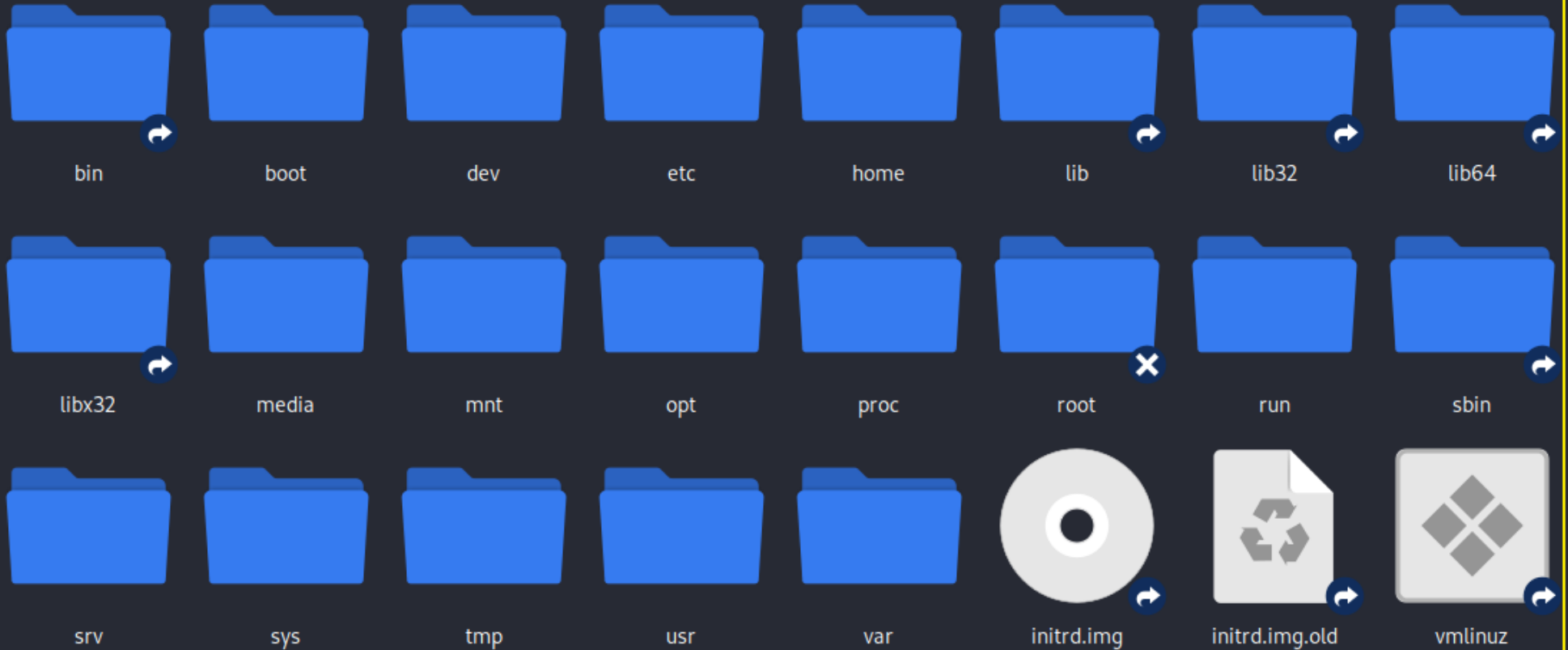
📁 Pictures

📁 Videos

📁 Downloads

NETWORK

🌐 Browse Network



vmlinuz.old

Show Linux file system from root /

Check where you are (impact all the commands you entered)

```
kali@kali: ~ 92x38

(kali㉿kali)-[~]
$ pwd          show current working directory
/home/kali

(kali㉿kali)-[~]
$ ls /         "/" mean root
bin      etc          initrd.img.old  lib64          media  proc  sbin  tmp  vmlinuz
boot     home            lib             libx32         mnt    root  srv   usr  vmlinuz.old
dev      initrd.img      lib32           lost+found     opt    run   sys   var
```

Clear screen

```
(kali㉿kali)-[~]
$ clear
```

Show details of a file/directory

```
(kali㉿kali)-[~]
```

```
$ ls -l /
```

```
total 68
```

```
lrwxrwxrwx  1 root root    7 Nov 17  2020 bin -> usr/bin
```

```
drwxr-xr-x  3 root root 4096 Nov 17  2020 boot
```

```
drwxr-xr-x 17 root root 3360 Jun 27 15:41 dev
```

```
drwxr-xr-x 157 root root 12288 Jun 27 15:29 etc
```

```
drwxr-xr-x  3 root root 4096 Nov 17  2020 home
```

```
lrwxrwxrwx  1 root root   33 Nov 17  2020 initrd.img -> boot/initrd.img-5.9.0-kali1-amd64
```

```
lrwxrwxrwx  1 root root   33 Nov 17  2020 initrd.img.old -> boot/initrd.img-5.9.0-kali1-amd64
```

```
lrwxrwxrwx  1 root root    7 Nov 17  2020 lib -> usr/lib
```

```
lrwxrwxrwx  1 root root    9 Nov 17  2020 lib32 -> usr/lib32
```

```
lrwxrwxrwx  1 root root    9 Nov 17  2020 lib64 -> usr/lib64
```

```
lrwxrwxrwx  1 root root   10 Nov 17  2020 libx32 -> usr/libx32
```

```
drwx-----  2 root root 16384 Nov 17  2020 lost+found
```

Show Linux file system from root with *tree*

```
(kali㉿kali)-[~]  
$ tree -L 1 /  
/  
├── bin -> usr/bin  
├── boot  
├── dev  
├── etc  
├── home  
├── initrd.img -> boot/initrd.img-5.9.0-kali1-amd64  
├── initrd.img.old -> boot/initrd.img-5.9.0-kali1-amd64  
├── lib -> usr/lib  
├── lib32 -> usr/lib32  
├── lib64 -> usr/lib64  
├── libx32 -> usr/libx32  
├── lost+found  
├── media  
├── mnt  
├── opt  
├── proc
```

only show Level 1

Remember use following
command for help
`tree --help`

Show *current working directory* (*pwd*) with *tree*

```
(kali㉿kali)-[~]  
$ pwd  
/home/kali  
  
(kali㉿kali)-[~]  
$ tree -L 1  
.  
├── Desktop  
├── Documents  
├── Downloads  
├── hacking_case  
├── Marvic_air  
├── Music  
├── Pictures  
├── pixel3  
├── Public  
├── Templates  
├── UB-730-Tools  
└── Videos  
  
12 directories, 0 files
```

```
(kali㉿kali)-[~]  
$ tree -L 2 | more  
.  
├── Desktop  
│   └── razorsql.desktop  
├── Documents  
├── Downloads  
│   ├── JLECmd.zip  
│   ├── razorsql  
│   ├── razorsql9_3_3_linux_x64.zip  
│   ├── RegRipper30-apt-git-Install.sh  
│   └── wine-mono-5.0.0-x86.msi  
└── hacking_case  
    ├── ip.txt  
    ├── mac.txt  
    ├── mirc.ini  
    ├── NTUSER_Evil.DAT  
    ├── RegRipper2.8  
    └── SAM
```

Linux commands are files!

Display `ls` command under `/bin`

```
(kali㉿kali)-[~]  
$ ls -l /bin/ls  
-rwxr-xr-x 1 root root 147176 Sep 24 2020 /bin/ls ←  
  
(kali㉿kali)-[~]  
$
```

Why can `/s` be executed in any folder? => Need to understand path (see next slide)

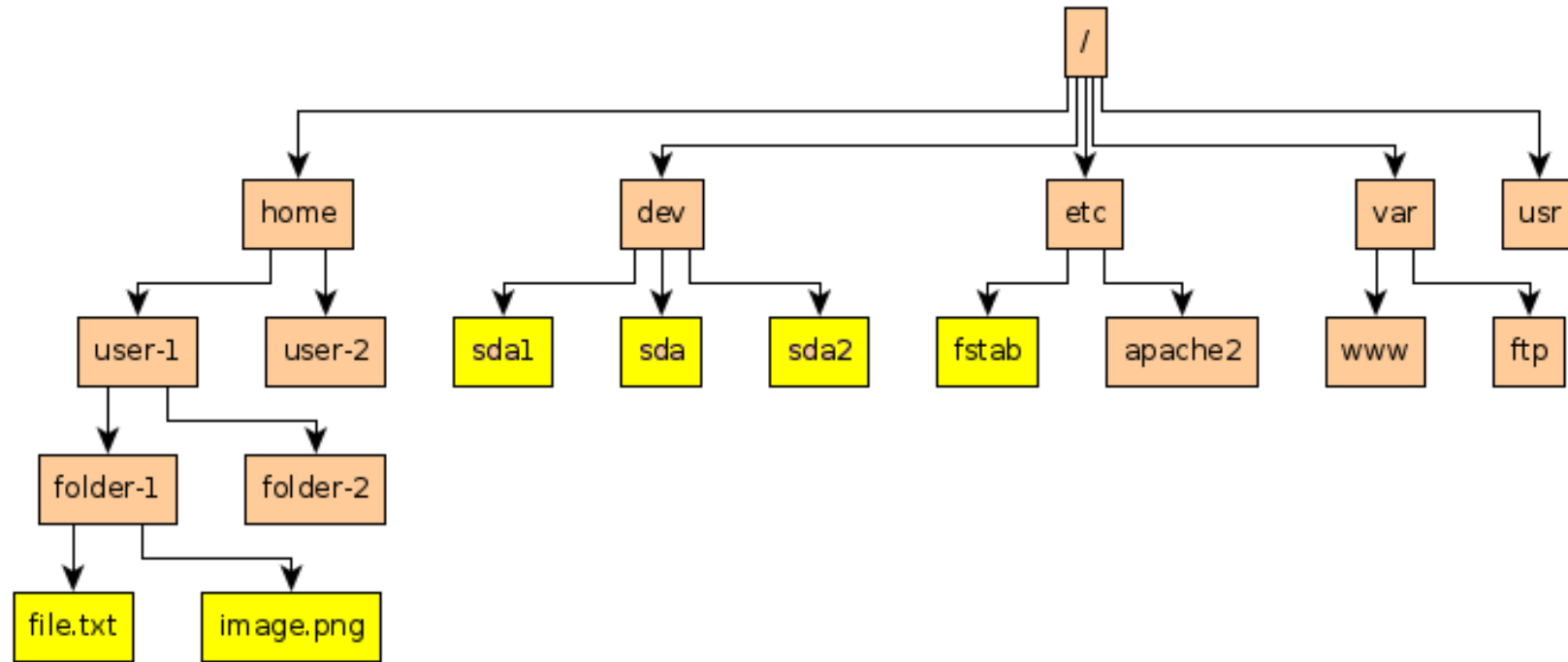
Path and Path variable

What is path in a file system?



- A path is a **hierarchical representation of the location** (address) of a file or directory within the file system's directory structure.
- A path describes the route or sequence of directories (folders) you must navigate through to reach a specific file or directory.
- Paths are used to uniquely identify and access files and directories on a computer or storage device.

Example of paths



- Absolute path to *file.txt*: `/home/user-1/folder-1/file.txt`
- Relative path to *file.txt* (Under *folder-2*): `../folder-1/file.txt`

Switch to the parent path using relative path

```
kali@kali: /home 82x25
(kaliⓈkali)-[~]
$ pwd
/home/kali ←

(kaliⓈkali)-[~]
$ cd ..      .. means parent folder
(kaliⓈkali)-[/home]
$ pwd
/home ←
```

Path variable (**\$PATH**)

```
kali@kali: ~ 82x25
(kali@kali)-[~]
$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/local/games:/usr/games
```

- How does OS execute a command, e.g., `ls`?
 - search the command in the current directory
 - if can't find it, search for the command in each path defined in the path variable
 - if can't find it, OS throws the *command not found* error message

```
(kali@kali)-[~]
$ ll
zsh: command not found: ll
```

Adding a new path (**export**)

```
(kali㉿kali)-[~]  
$ export PATH=/some/new/path:$PATH  
  
(kali㉿kali)-[~]  
$ echo $PATH  
/some/new/path:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/  
local/games:/usr/games
```

new added path



```
setx path "%path%;C:\Program Files\Java\jdk1.8.0_202\bin" -m
```

Persisting the new path (**.zshrc**)

```
(kali㉿kali)-[~]  
$ ls -a  
.  
..  
.bash_history  
.bash_logout  
.bashrc  
.bashrc.original  
.cache  
.config  
Desktop  
.dmrc  
Documents  
Downloads  
.face  
.face.icon  
.gnupg  
hacking_case  
.ICEauthority  
.java  
.local  
Marvic_air  
.mozilla  
Music  
Pictures  
pixel3  
.profile  
Public  
.razorsql  
.rs  
.sqlite_history  
Templates  
UB-730-Tools  
.vboxclient-clipboard.pid  
.vboxclient-display-svga-x11.pid  
.vboxclient-draganddrop.pid  
.vboxclient-seamless.pid  
Videos  
.viminfo  
.wget-hsts  
.Xauthority  
.xsession-errors  
.xsession-errors.old  
.zsh_aliases  
.zsh_history  
.zshrc
```

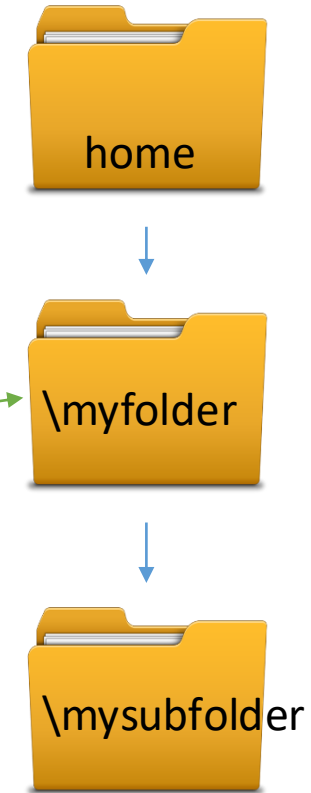
← add to this file

we discuss the details later

Create folders and files

Create folders (**mkdir**)

```
root@kali:~# mkdir myfolder
root@kali:~# cd myfolder/
root@kali:~/myfolder# mkdir mysubfolder
root@kali:~/myfolder# ls
mysubfolder
root@kali:~/myfolder#
```



Delete folders (**rmdir**)

```
root@kali:~/myfolder# ls
mysubfolder
root@kali:~/myfolder# rmdir mysubfolder/
root@kali:~/myfolder# ls
root@kali:~/myfolder#
```

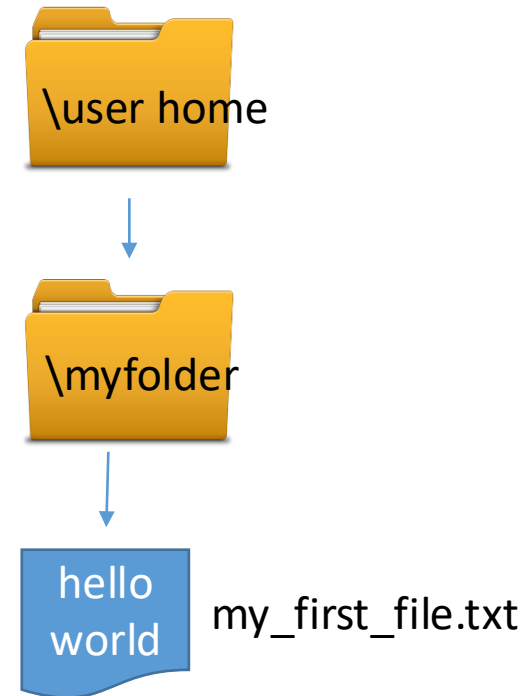


Create a new text file (**echo**)

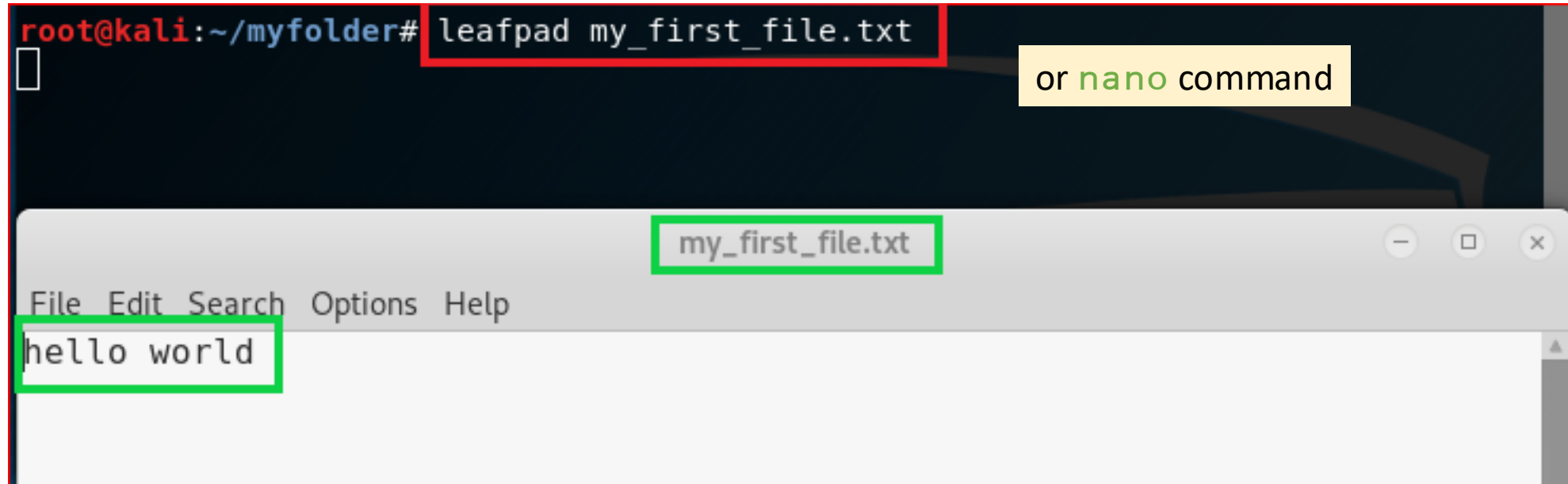
```
root@kali:~/myfolder# echo "hello world" > my_first_file.txt
root@kali:~/myfolder# ls
my_first_file.txt
root@kali:~/myfolder#
```

Show the content of the file

```
root@kali:~/myfolder# cat my_first_file.txt
hello world
root@kali:~/myfolder#
```



Create a file (**Leafpad**)



The screenshot shows a terminal window at the top with the command `leafpad my_first_file.txt` entered. Below the terminal is a window titled `my_first_file.txt` with a menu bar containing `File`, `Edit`, `Search`, `Options`, and `Help`. The text `hello world` is visible in the editor's text area. A yellow callout box points to the terminal command with the text `or nano command`.

```
root@kali:~/myfolder# leafpad my_first_file.txt
```

or nano command

my_first_file.txt

File Edit Search Options Help

hello world

File Copy & Deletion

Copy a file (**cp**)

```
root@kali:~/myfolder# cp my_first_file.txt my_first_file_dup.txt
root@kali:~/myfolder# ls
my_first_file_dup.txt  my_first_file.txt
root@kali:~/myfolder#
```



copy

Copy a file to a different location

```
root@kali:~/myfolder# cp my_first_file.txt /var/www/html
root@kali:~/myfolder# ls /var/www/html/my_first_file.txt
/var/www/html/my_first_file.txt
root@kali:~/myfolder#
```

How to copy a folder? **-r** recursive

- Basic syntax: **cp -r** source_folder destination_folder
- Example: **cp -r** ~/myfolder ~/myfolder_copy

Remove a file (**rm**)

```
root@kali:~/myfolder# ls
my_first_file_dup.txt  my_first_file.txt
root@kali:~/myfolder# rm my_first_file_dup.txt
root@kali:~/myfolder# ls
my_first_file.txt
root@kali:~/myfolder#
```

Rename a file (**mv**)

```
root@kali:~/myfolder# ls
my_first_file.txt
root@kali:~/myfolder# mv my_first_file.txt renamed_file.txt
root@kali:~/myfolder# ls
renamed_file.txt
root@kali:~/myfolder#
```



Search for information

Search for a string in a text file (**grep**)

grep search

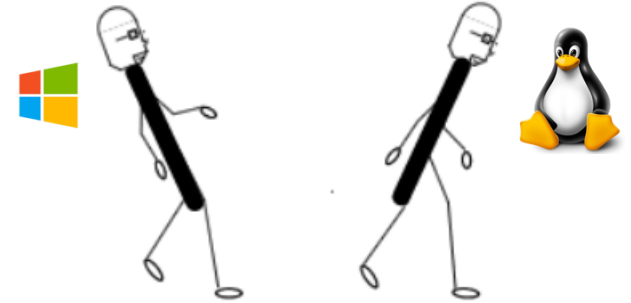
```
(kali㉿kali)-[~/myfolder]
$ ls
my_first_file.txt ←

(kali㉿kali)-[~/myfolder]
$ cat my_first_file.txt
hello world ←

(kali㉿kali)-[~/myfolder]
$ grep "hello" my_first_file.txt
hello world ←
```

Show line number while displaying the output

```
(kali㉿kali)-[~/myfolder]
$ grep -n "hello" my_first_file.txt
1:hello world ←
```



backslash

forward slash



type/findstr /c:"error" log.txt

Search for file names that contains the given string/pattern

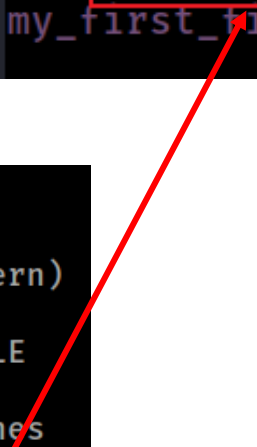
grep help

```
kali@kali: ~/myfolder 81x48
(kali@kali)-[~/myfolder]
$ grep --help
Usage: grep [OPTION]... PATTERNS [FILE]...
Search for PATTERNS in each FILE.
Example: grep -i 'hello world' menu.h main.c
PATTERNS can contain multiple patterns separated by newlines.

-r, --recursive           like --directories=recurse
-R, --dereference-recursive likewise, but follow all symlinks
--include=GLOB            search only files that match GLOB (a file pattern)
--exclude=GLOB            skip files that match GLOB
--exclude-from=FILE       skip files that match any file pattern from FILE
--exclude-dir=GLOB        skip directories that match GLOB
-L, --files-without-match print only names of FILES with no selected lines
-l, --files-with-matches  print only names of FILES with selected lines
-c, --count               print only a count of selected lines per FILE
```

Search for file contains "hello"

```
kali@kali: ~/myfolder
(kali@kali)-[~/myfolder]
$ grep -l "hello" *
my_first_file.txt
```



Networking

Test network connection using domain name

```
root@kali:~# ping google.com
PING google.com (216.58.219.206) 56(84) bytes of data.
64 bytes from lga25s40-in-f14.1e100.net (216.58.219.206): icmp_seq=1 ttl=56 time=10.9 ms
64 bytes from lga25s40-in-f14.1e100.net (216.58.219.206): icmp_seq=2 ttl=56 time=11.4 ms
64 bytes from lga25s40-in-f14.1e100.net (216.58.219.206): icmp_seq=3 ttl=56 time=12.9 ms
64 bytes from lga25s40-in-f14.1e100.net (216.58.219.206): icmp_seq=4 ttl=56 time=10.10 ms
64 bytes from lga25s40-in-f14.1e100.net (216.58.219.206): icmp_seq=5 ttl=56 time=12.2 ms
```

Test network connection using IP

```
root@kali:~# ping 216.58.219.206
PING 216.58.219.206 (216.58.219.206) 56(84) bytes of data.
64 bytes from 216.58.219.206: icmp_seq=1 ttl=56 time=11.1 ms
64 bytes from 216.58.219.206: icmp_seq=2 ttl=56 time=12.3 ms
64 bytes from 216.58.219.206: icmp_seq=3 ttl=56 time=12.4 ms
64 bytes from 216.58.219.206: icmp_seq=4 ttl=56 time=15.3 ms
^C
```

List all open TCP ports

-l, --listening display listening server sockets
-n, --numeric port number (don't resolve names)
-t, --tcp

```
student@kalit01: ~ 80x24
(student@kalit01)-[~]
$ netstat -lnt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      0 127.0.0.1:5432          0.0.0.0:*               LISTEN
tcp      0      0 0.0.0.0:80              0.0.0.0:*               LISTEN
tcp      0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp6     0      0 :::3389                  :::*                     LISTEN
tcp6     0      0 :::22                    :::*                     LISTEN
tcp6     0      0 :::1:3350                :::*                     LISTEN
tcp6     0      0 :::1:5432                 :::*                     LISTEN
```

ssh (22) port is listening

Open port 21 and verify the port is listening

```
(student@kalit01)-[~]
```

```
$ netstat -lnt
```

Active Internet connections (only servers)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	127.0.0.1:5432	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:80	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:22	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:21	0.0.0.0:*	LISTEN
tcp6	0	0	:::3389	:::*	LISTEN
tcp6	0	0	:::22	:::*	LISTEN
tcp6	0	0	:::1:3350	:::*	LISTEN
tcp6	0	0	:::1:5432	:::*	LISTEN

2. listening

```
(student@kalit01)-[~]
```

```
$
```

```
(student@kalit01)-[~]
```

```
$ nc -lp 21
```

1. listen to port 21

nmap localhost -p 21

Download a file (**wget**)

```
root@kali: ~/myfolder
File Edit View Search Terminal Help
root@kali:~# cd myfolder/
root@kali:~/myfolder# wget https://pbs.twimg.com/media/DuILzQXcAAkFMV.jpg
--2018-12-16 22:49:03-- https://pbs.twimg.com/media/DuILzQXcAAkFMV.jpg
Resolving pbs.twimg.com (pbs.twimg.com)... 72.21.91.70, 2606:2800:220:1410:4
89:141e:20bb:12f6
Connecting to pbs.twimg.com (pbs.twimg.com)|72.21.91.70|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 109010 (106K) [image/jpeg]
Saving to: 'DuILzQXcAAkFMV.jpg'

DuILzQXcAAkFMV.jpg  100%[=====] 106.46K  --.-KB/s
in 0.1s

2018-12-16 22:49:03 (894 KB/s) - 'DuILzQXcAAkFMV.jpg' saved [109010/109010]

root@kali:~/myfolder# ls
DuILzQXcAAkFMV.jpg  renamed file.txt
root@kali:~/myfolder# display DuILzQXcAAkFMV.jpg
```



<https://pbs.twimg.com/media/DuILzQXcAAkFMV.jpg>

Create a script file

Create a simple script file

Create a script using leafpad

```
(kali㉿kali)-[~/myfolder]
$ leafpad myFirstScript.sh
#!/bin/sh
echo "Hello World" > myFile.txt
ls myFile.txt -l
```

shebang: indicate the interpreter that should be used to execute the script or program that follows

Check permission of files

```
(kali㉿kali)-[~/myfolder]
$ ls -l
total 8
-rw-r--r-- 1 kali kali 12 Jun 27 22:19 my_first_file.txt
-rw-r--r-- 1 kali kali 48 Jun 27 23:01 myFirstScript.sh
```

```
# ls -l file
-rw-r--r-- 1 root root 0 Nov 19 23:49 file
```

File type

Owner (rw-) Group (r- -) Other (r - -)

r = Readable
w = Writeable
x = Executable
- = Denied

Linux Permissions

u g o
754

	r	w	x	r	w	x	r	w	x
access	r	w	x	r	w	x	r	w	x
binary	4	2	1	4	2	1	4	2	1
enabled	1	1	1	1	0	1	1	0	0
result	4	2	1	4	0	1	4	0	0
total	7			5			4		

```
root@kali: ~/myfolder
root@kali: ~/myfolder 80x24
root@kali:~/myfolder# chmod 777 myFirstScript.sh
root@kali:~/myfolder# ls myFirstScript.sh -l
-rwxrwxrwx 1 root root 49 Mar 20 11:13 myFirstScript.sh
root@kali:~/myfolder#
```

```
root@kali: ~/myfolder
root@kali: ~/myfolder 80x24
root@kali:~/myfolder# ./myFirstScript.sh
-rw-r--r-- 1 root root 12 Mar 20 11:14 myFile.txt
root@kali:~/myfolder#
```

Execute a script file in a different folder

```
(kali㉿kali)-[~/myfolder]
$ cd ..

(kali㉿kali)-[~]
$ pwd
/home/kali

(kali㉿kali)-[~]
$ ./myfolder/myFirstScript.sh
-rw-r--r-- 1 kali kali 12 Jun 28 11:06 myFile.txt
```

MUST include path

Add a path to the path variable

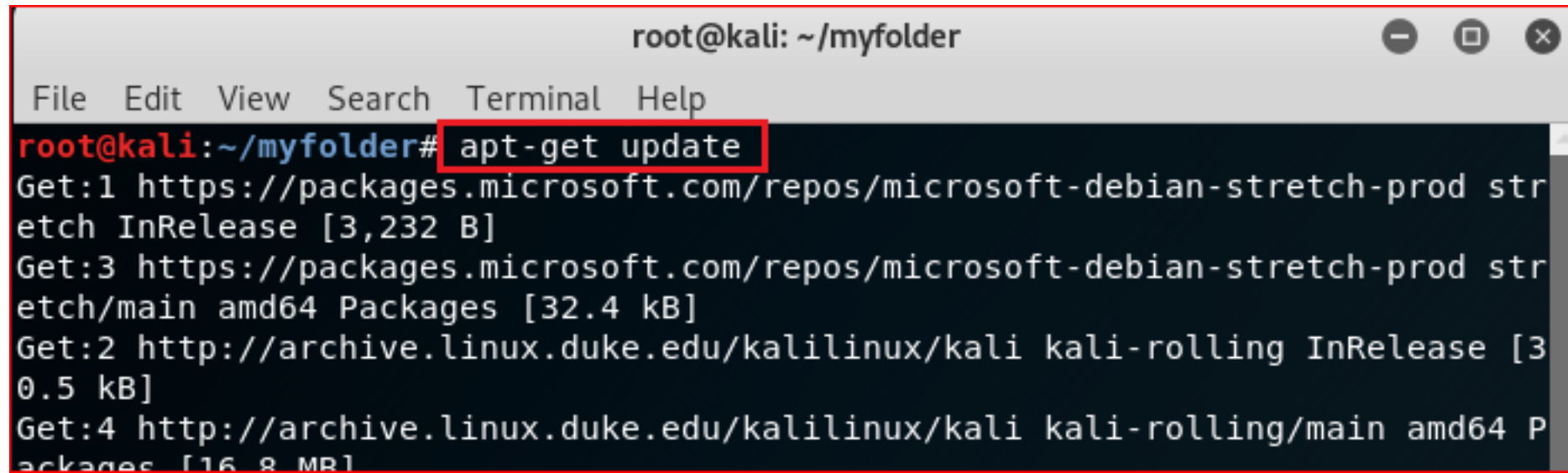
```
(kali㉿kali)-[~]  
$ pwd  
/home/kali  
  
(kali㉿kali)-[~]  
$ ls ./myfolder -l  
total 12  
-rw-r--r-- 1 kali kali 12 Jun 27 23:06 myFile.txt  
-rw-r--r-- 1 kali kali 12 Jun 27 22:19 my_first_file.txt  
-rwxrwxrwx 1 kali kali 48 Jun 27 23:01 myFirstScript.sh  
  
(kali㉿kali)-[~]  
$ echo $PATH  
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/local/games:/usr/games  
  
(kali㉿kali)-[~]  
$ export PATH=/home/kali/myfolder:$PATH  
  
(kali㉿kali)-[~]  
$ echo $PATH  
/home/kali/myfolder:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/local/games:/usr/games  
  
(kali㉿kali)-[~]  
$ myFirstScript.sh  
-rw-r--r-- 1 kali kali 12 Jun 28 11:18 myFile.txt
```

executed without specifying path

Update/Install software

(optional)

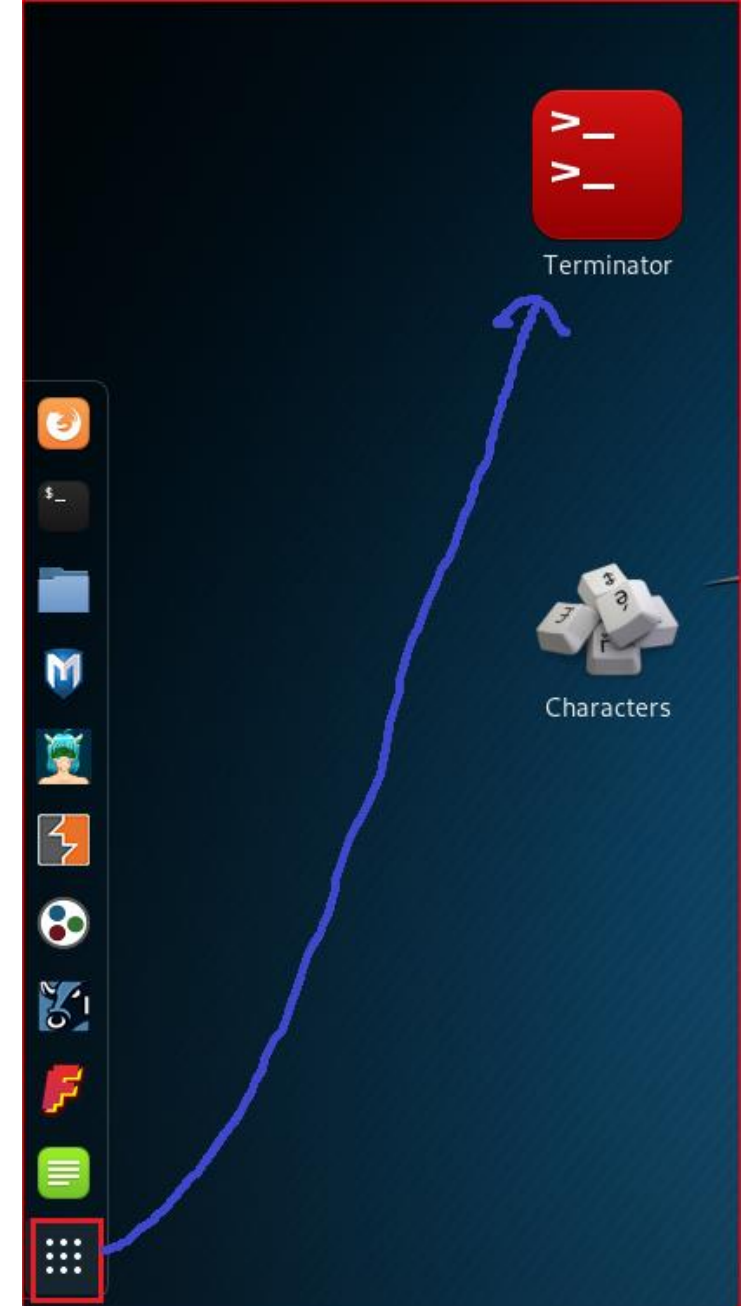
Update software

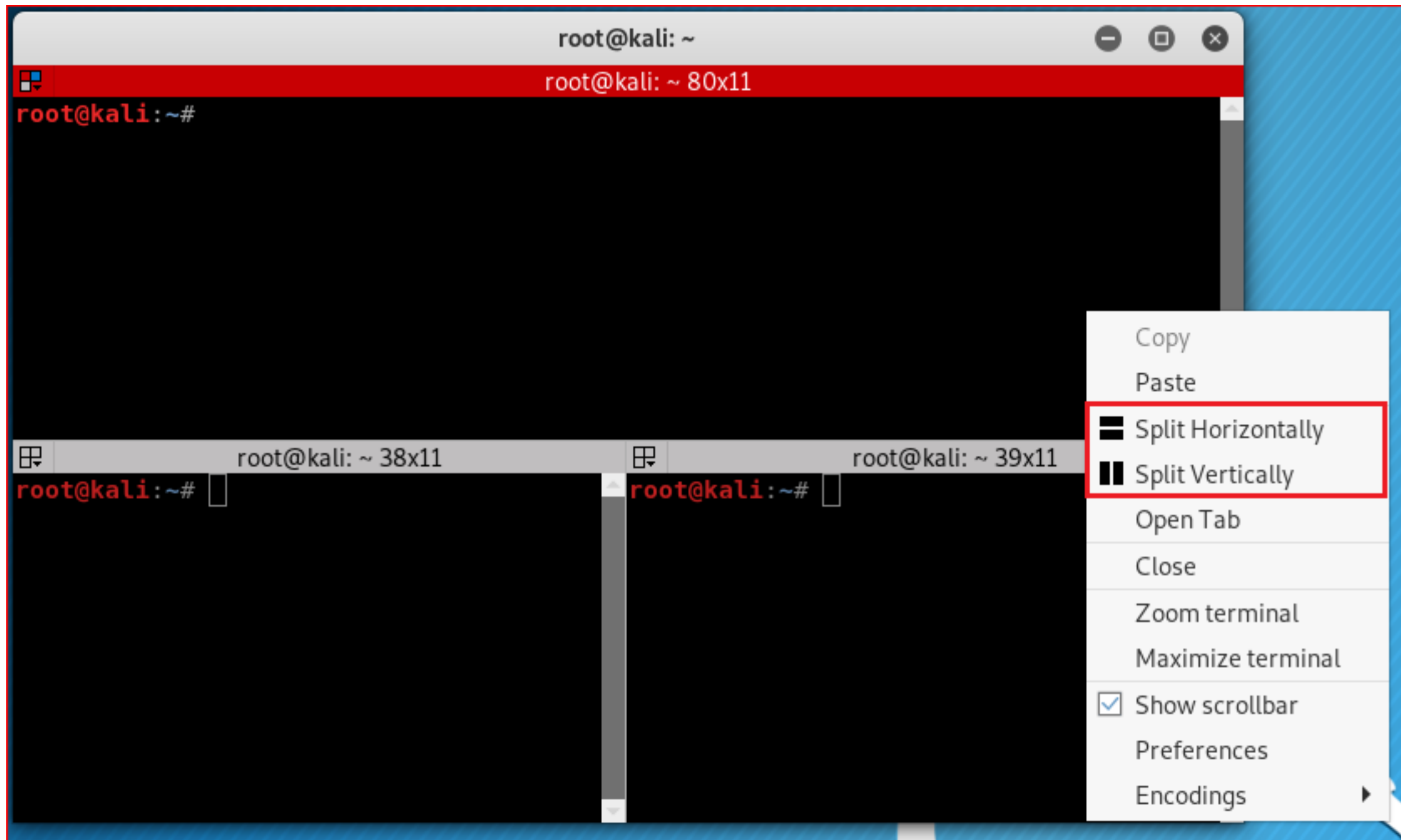


```
root@kali: ~/myfolder
File Edit View Search Terminal Help
root@kali:~/myfolder# apt-get update
Get:1 https://packages.microsoft.com/repos/microsoft-debian-stretch-prod stretch InRelease [3,232 B]
Get:3 https://packages.microsoft.com/repos/microsoft-debian-stretch-prod stretch/main amd64 Packages [32.4 kB]
Get:2 http://archive.linux.duke.edu/kalilinux/kali kali-rolling InRelease [30.5 kB]
Get:4 http://archive.linux.duke.edu/kalilinux/kali kali-rolling/main amd64 Packages [16.8 MB]
```

Install terminator

```
root@kali:~/myfolder# apt-get install terminator
Reading package lists... Done
Building dependency tree
Reading state information... Done
terminator is already the newest version (1.91-1).
```





Zip/unzip files

```
root@kali:~/myfolder# ls
DulILzQXcAAkFMV.jpg  renamed_file.txt
root@kali:~/myfolder# zip zipped_image.zip DulILzQXcAAkFMV.jpg
  adding: DulILzQXcAAkFMV.jpg (deflated 1%)
root@kali:~/myfolder# ls
DulILzQXcAAkFMV.jpg  renamed_file.txt  zipped_image.zip
root@kali:~/myfolder# rm DulILzQXcAAkFMV.jpg
root@kali:~/myfolder# ls
renamed_file.txt  zipped_image.zip
root@kali:~/myfolder#
```

root@kali: ~/myfolder 80x11

```
root@kali:~/myfolder# unzip zipped_image.zip
Archive:  zipped_image.zip
  inflating: DulILzQXcAAkFMV.jpg
root@kali:~/myfolder# ls
DulILzQXcAAkFMV.jpg  renamed_file.txt  zipped_image.zip
root@kali:~/myfolder#
```

Compress/decompress directory (tar)

compress a folder

```
(wxu@kalit01)-[~]  
$ ls -l myfolder  
total 4  
-rw-r--r-- 1 wxu student 6 Jan 24 15:56 mytext.txt  
  
(wxu@kalit01)-[~]  
$ tar -czvf archive.tar.gz myfolder  
myfolder/  
myfolder/mytext.txt  
  
(wxu@kalit01)-[~]  
$ ls archive.tar.gz -l  
-rw-r--r-- 1 wxu student 168 Jan 24 16:01 archive.tar.gz
```

- c: Create a new archive.
- z: Compress the archive using gzip (creates a .tar.gz file).
- v: Verbose mode (shows the files being processed).
- f: Specify the archive file name.

compress a folder

```
(wxu@kalit01)-[~]  
$ tar -xzvf archive.tar.gz  
myfolder/  
myfolder/mytext.txt
```

-x: Extract files.

-z: Decompress using gzip.

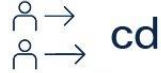
Linux Basic Commands

Every User Should Know



pwd

To find out the path of the current working directory



cd

To navigate through the Linux files and directories



ls

Is used to view the contents of a directory



cat

Is used to create a new file



cp

To copy files from the current directory to a different directory



mv

The command is to move files



mkdir

Use mkdir command to make a new directory



rmdir

The rm command is used to delete directories and the contents within them



locate

You can use this command to locate a file, just like the search command in Windows



sudo

This command enables you to perform tasks that require administrative or root permissions



head

The head command is used to view the first lines of any text

Some Basic Linux Commands

by @SecurityGuill



FILE COMMANDS

- **ls** = directory listing
- **ls -al** = formatted listing with hidden files
- **cd dir** = change directory to dir
- **pwd** = show current directory
- **mkdir dir** = create directory dir
- **rm file** = delete file
- **rm -r dir** = delete directory dir
- **rm -f file** = force remove file
- **rm -rf dir** = force remove directory
- **cp file1 file2** = copy file1 to file2
- **mv file1 file2** = rename file1 to file2
- **ln -s file link** = create symbolic link 'link' to file
- **touch file** = create or update file
- **cat > file** = place standard input into file
- **more file** = output the contents of the file
- **less file** = output the contents of the file
- **head file** = output first 10 lines of file
- **tail file** = output last 10 lines of file
- **tail -f file** = output contents of file as it grows



NETWORK

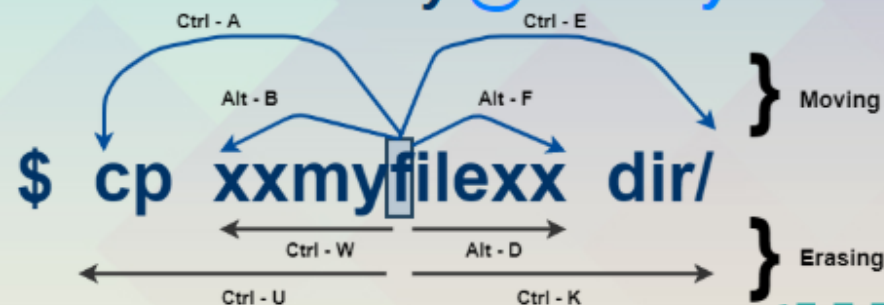


- **ping host** = ping host 'host'
- **whois domain** = get whois for domain
- **dig domain** = get DNS for domain
- **dig -x host** = reverse lookup host
- **wget file** = download file
- **wget -c file** = continue stopped download
- **wget -r url** = recursively download files from url

SYSTEM INFO



- **date** = show current date/time
- **cal** = show this month's calendar
- **uptime** = show uptime
- **w** = display who is online
- **whoami** = who are you logged in as
- **uname -a** = show kernel config
- **cat /proc/cpuinfo** = cpu info
- **cat /proc/meminfo** = memory info
- **man command** = show manual for command
- **df** = show disk usage
- **du** = show directory space usage
- **du -sh** = human readable size in GB
- **free** = show memory & swap usage
- **whereis app** = show possible locations of app
- **which app** = show which app will be run by default



SEARCHING

- **grep pattern files** = search for pattern in files
- **grep -r pattern dir** = search recursively for pattern in dir
- **command | grep pattern** = search for pattern in the output of command
- **locate file** = find all instances of file

PROCESS MANAGEMENT

- **ps** = display currently active processes
- **ps aux** = ps with a lot of detail
- **kill pid** = kill process with pid 'pid'
- **killall proc** = kill all processes named proc
- **bg** = lists stopped/background jobs
- **fg** = bring most recent job to foreground
- **fg n** = brings job n to foreground

FILE PERMISSIONS

- **chmod octal file** = change permission of file (4:read / 2:write / 1:execute)
Order: owner/group/world
Eg: **chmod 755 file** = read write for owner, read execute for group/world



COMPRESSION

- **tar cf file.tar files** = tar files into file.tar
- **tar xf file.tar** = untar into current directory
- **tar tf file.tar** = show contents of archive

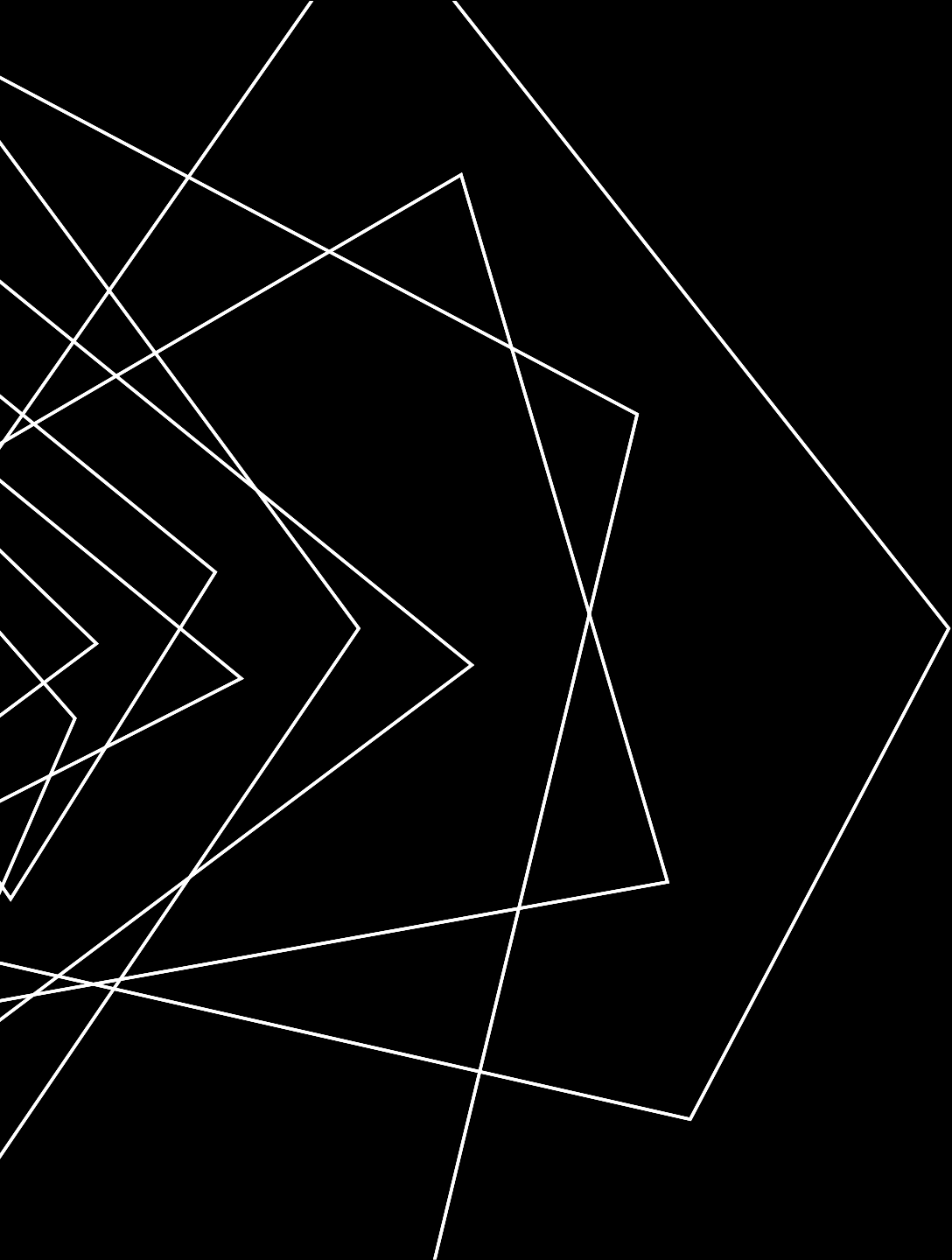
SSH



- **ssh user@host** = connect to host
- **ssh -p port user@host** = connect using port p
- **ssh -D port user@host** = connect & use bind port



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Thank you

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